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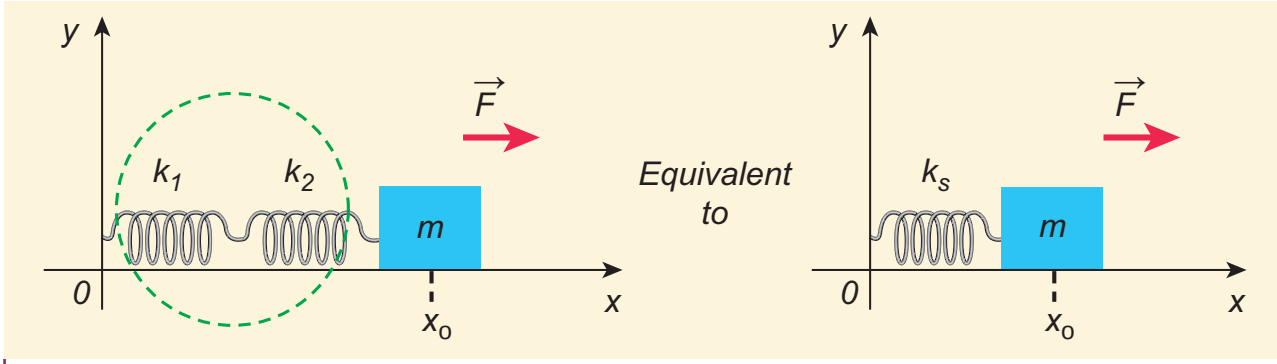


Figure 10.18 Effective spring constant in series connection

$$F = -k_s (x_1 + x_2) \Rightarrow x_1 + x_2 = -\frac{F}{k_s} \quad (10.38)$$

For springs in series connection

$$-k_1 x_1 = -k_2 x_2 = F$$

$$\Rightarrow x_1 = -\frac{F}{k_1} \text{ and } x_2 = -\frac{F}{k_2} \quad (10.39)$$

Therefore, substituting equation (10.39) in equation (10.38), the *effective spring constant* can be calculated as

$$-\frac{F}{k_1} - \frac{F}{k_2} = -\frac{F}{k_s}$$

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2}$$

Or

$$k_s = \frac{k_1 k_2}{k_1 + k_2} \text{ Nm}^{-1} \quad (10.40)$$

Suppose we have n springs connected in series, the effective spring constant in series is

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \ldots + \frac{1}{k_n} = \sum_{i=1}^{n} \frac{1}{k_i} \quad (10.41)$$

If all spring constants are identical i.e., $k_1 = k_2 = \ldots = k_n = k$ then

$$\frac{1}{k_s} = \frac{n}{k} \Rightarrow k_s = \frac{k}{n} \quad (10.42)$$

This means that the effective spring constant reduces by the factor n. Hence, for springs in series connection, the effective spring constant is lesser than the individual spring constants.

From equation (10.39), we have,

$$k_1 x_1 = k_2 x_2$$

Then the ratio of compressed distance or elongated distance x_1 and x_2 is

$$\frac{x_2}{x_1} = \frac{k_1}{k_2} \quad (10.43)$$

The elastic potential energy stored in first and second springs are $V_1 = \frac{1}{2} k_1 x_1^2$ and $V_2 = \frac{1}{2} k_2 x_2^2$ respectively. Then, their ratio is

$$\frac{V_1}{V_2} = \frac{\frac{1}{2} k_1 x_1^2}{\frac{1}{2} k_2 x_2^2} = \frac{k_1}{k_2} \left(\frac{x_1}{x_2} \right)^2 = \frac{k_2}{k_1} \quad (10.44)$$

Note

The reciprocal of stiffness constant is called flexibility constant or compliance, denoted by C. It is measured in m N^{-1}

If n springs are connected in series :

$$\text{net compliance } C_s = \sum_{i=1}^{n} C_i$$

If n springs are connected in parallel :

$$\frac{1}{C_p} = \sum_{i=1}^{n} \frac{1}{C_i}$$